Career Simulation Platform

Complete Product, Research, Content, Safety, and Technical Architecture

1. Product Vision

The platform should help users understand:

- What a career actually feels like
- What professionals do during ordinary work, not only dramatic moments
- Which parts of the work they enjoy or dislike
- Which skills they currently demonstrate
- Which skills they may need to develop
- Whether they prefer the pace, environment, responsibilities, and social demands
- What realistic education, credentialing, financial, and employment pathways lead into the career

The product should not behave like:

- A personality quiz with graphics
- An unrestricted AI role-play chatbot
- A career-matching test that declares what someone should become
- A video game that prioritizes entertainment over occupational accuracy
- A single-session assessment pretending to make precise psychological judgments

The product should be:

> A scalable, bounded career world where users explore workplaces, operate realistic work systems, interact with people, manage authentic demands, experience consequences, and build a longitudinal evidence record about what kinds of work they may want to explore further.

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2. Core Product Model

The experience should simulate four layers of work simultaneously.

A. The environment

Where the work occurs:

- Office
- Classroom

- Hospital unit
- Courtroom
- Construction site
- Workshop
- Laboratory
- Remote workspace
- Client location

B. Professional systems

What professionals actually use:

- Email
- Messaging
- Calendars
- Documents
- Dashboards
- Records
- Equipment
- Ticket systems
- Reports
- Internal policies
- Knowledge bases

C. Human experience

What the work feels like:

- Focus
- Interruption
- Ambiguity
- Responsibility
- Repetition
- Conflict
- Collaboration
- Emotional demands
- Cognitive load
- Social exposure
- Physical demands
- Slow progress
- Uncertainty
- Satisfaction from completing careful work

D. Career pathway reality

What pursuing the career requires:

- Education
- Credentials
- Licensure
- Training time
- Cost
- Entry barriers
- Working conditions
- Typical progression
- Compensation ranges
- Geographic constraints
- Job-market considerations

Why all four layers are necessary

A simulation that only models the physical workplace becomes a shallow game.

A simulation that only presents decisions becomes a disguised quiz.

A simulation that ignores training and employment realities can make a career feel attractive without helping the user understand whether pursuing it is realistic.

The platform must connect:

```
What the work feels like
+
How the user responded
+
What pursuing the career requires
```

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3. Primary Design Principles

Principle 1: Simulate occupational truth, not excitement

The purpose is not to make every career dramatic.

Some careers involve:

- Constant interruption
- Fast triage

- High emotional pressure

Others involve:

- Long concentration periods
- Repetitive but satisfying work
- Slow investigation
- Quiet organization
- Precision
- Predictable routines

A librarian, actuary, laboratory technician, accountant, and emergency nurse should not be forced through the same narrative arc.

Reason

If every career uses emergencies and escalating tension, the system introduces an excitement bias. High-stimulation careers appear more meaningful, while calm careers appear less immersive even when calmness is their authentic appeal.

The simulation must treat preferences for:

- Quiet focus
- Routine
- Predictability
- Low social exposure
- Deliberate pace

as valid career signals.

Principle 2: Simulate pressure, not polygons

Immersion should primarily come from:

- Realistic choices
- Competing demands
- Persistent consequences
- Social reactions
- Honest tedium
- Environmental context
- Control over attention

High-end visuals should be added only when they materially improve career understanding.

Psychological reason

Users feel present when their decisions matter and the world reacts consistently. Visual quality cannot compensate for arbitrary consequences or inaccurate work.

Technical reason

Detailed 3D environments greatly increase:

- Art requirements
- Animation work
- Device demands
- Testing complexity
- Accessibility challenges
- Development time per career

A lightweight navigable workplace offers spatial presence without making every career a separate game-development project.

Principle 3: Exploration, not diagnosis

The platform should never claim:

- “You are definitely suited for this career.”
- “You are bad at communication.”
- “This simulation proves you should become a nurse.”
- “Your career compatibility is 87%.”

It should instead say:

- “In this session, you repeatedly sought additional evidence before acting.”
- “You completed the task successfully but reported that the interruption level was draining.”
- “This is an early observation based on limited evidence.”
- “A second simulation would help determine whether this pattern repeats.”

Reason

One simulation cannot establish a stable psychological trait or predict long-term career success. Presenting exploratory evidence as diagnosis would create false precision and may influence major decisions without adequate support.

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4. Research and Validation Workstream

Validation must begin alongside product development, not after the platform is built.

The platform has two separate claims to test:

Claim A: Experiential validity

> Does the simulation feel meaningfully representative of the career?

Claim B: Decision usefulness

> Does completing the simulation improve the quality of a user's career exploration or next decision?

These are different.

A simulation can feel convincing while presenting an inaccurate version of the occupation.

4.1 Establish comparison baselines

The platform should be compared against:

- Traditional career descriptions
- Existing career-interest questionnaires
- Occupational databases
- Informational interviews
- Job-shadowing where available
- Standard career-counseling activities

Reason

Without a baseline, a positive user reaction does not demonstrate that the simulation provides better understanding than existing methods.

4.2 Validation stages

Stage 1: Content validity

Professionals in the field review whether the simulation accurately represents:

- Tasks
- Tools
- Pace
- Decisions
- Social relationships
- Repetitive work

- Entry-level responsibility
- Consequences
- Common misconceptions

Stage 2: Perceived realism

Users and practitioners separately rate:

- Whether the simulation felt coherent
- Whether it represented ordinary work
- Whether important parts were missing
- Whether it overemphasized dramatic situations

Practitioner ratings should carry more weight for occupational accuracy.

Stage 3: Decision usefulness

Test whether users can more accurately explain:

- What the occupation involves
- Which parts they would enjoy
- Which parts they would dislike
- What training is required
- What additional exploration they should pursue

Stage 4: Convergent evidence

Compare simulation observations against:

- Established interest measures
- Self-reported preferences
- Counselor observations
- Performance in related educational tasks

The goal is not exact agreement. Disagreement can be useful, but it should be understood.

Stage 5: Longitudinal validation

Where possible, examine whether patterns repeat across:

- Multiple sessions
- Different scenarios within one career
- Related careers
- Internships
- Job-shadowing
- Coursework
- Later career choices

Stage 6: Predictive research

Only after sufficient longitudinal evidence should the platform study whether simulation behavior relates to real-world performance.

Until then, it should not imply that success in a simulation predicts success in employment.

4.3 Research partners

The project should involve, where practical:

- Career-development researchers
- Industrial-organizational psychologists
- School counselors
- Accessibility specialists
- Occupational subject-matter experts
- Learning-science researchers

Why this is required

Software developers can reliably track actions, but they should not independently decide what those actions mean psychologically.

The product team can build the evidence engine. Qualified research partners should help define what conclusions the engine is permitted to draw.

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5. Experience Format

Selected approach

Build a lightweight navigable workplace in which users can:

- Move between locations
- Approach characters
- Open professional systems
- Inspect objects
- Decide what deserves attention
- Ask questions
- Delay, delegate, complete, or ignore tasks
- Observe how the environment changes over time

The first version may use:

- Illustrated scenes
- A room map
- Clickable locations
- Simple character animations
- 2.5D environments
- Lightweight 3D rooms

It does not need unrestricted open-world movement.

Why not text-only?

Text can represent complex decisions, but spatial interaction contributes to:

- Presence
- Environmental memory
- Attention management
- Workplace understanding
- User agency

Walking to a workstation and opening an incident dashboard feels more embodied than selecting “check logs” from a menu.

Why not full 3D immediately?

Full 3D creates large costs without proving that it improves career understanding.

The platform should first validate that:

- The occupational content is accurate
- Users interpret the simulation correctly
- The debrief is useful
- The engine generalizes across careers

Visual fidelity can increase after those foundations are established.

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6. Accessible Parallel Interaction Modes

The platform should support at least two interaction modes using the same simulation state.

Mode A: Spatial mode

Users navigate through rooms and interact with objects visually.

Mode B: Structured accessible mode

Users access the same world through:

- Clear location menus
- Keyboard navigation
- Screen-reader-compatible descriptions
- Text-based interaction panels
- Adjustable pacing
- Reduced visual motion

Both modes must produce the same available actions and consequences.

Why this matters

Difficulty using a game-like interface must not be mistaken for difficulty performing the simulated career.

A non-gamer may struggle to locate an object even though they understand the professional decision perfectly.

Separating navigation ability from occupational behavior reduces this confound.

Mandatory onboarding

Before any assessed simulation, users should complete a neutral tutorial that teaches:

- Movement
- Opening systems
- Speaking to NPCs
- Inspecting objects
- Managing tasks
- Pausing
- Accessibility controls

The tutorial should not produce career evidence.

Reason

Early mistakes may reflect unfamiliarity with the platform rather than authentic decision patterns.

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7. Scenario Architecture

The platform should not use one universal dramatic backbone.

Instead, career packs select from multiple scenario rhythms.

Scenario Type A: High-pressure operational

Suitable for:

- Emergency healthcare
- Cybersecurity operations
- Live event production
- Dispatch
- Incident management

Typical rhythm:

```
Initial responsibilities
→ Competing signals
→ Prioritization
→ Escalating operational risk
→ Coordination
→ Resolution and handoff
```

Scenario Type B: Quiet competence

Suitable for:

- Accounting
- Archival work
- Actuarial analysis
- Research
- Quality assurance
- Laboratory analysis

Typical rhythm:

```
Receive objective
→ Gather information
→ Perform focused work
→ Detect subtle inconsistency
```

```
→ Decide whether to investigate
→ Complete and document
```

No artificial emergency is required.

Scenario Type C: Social and relational

Suitable for:

- Teaching
- Counseling
- Management
- Human resources
- Sales
- Community work

Typical rhythm:

```
Build context
→ Interpret another person's needs
→ Communicate
→ Handle misunderstanding
→ Maintain boundaries
→ Follow up
```

Scenario Type D: Creative ambiguity

Suitable for:

- Design
- Writing
- Product development
- Marketing
- Architecture

Typical rhythm:

```
Receive incomplete brief
→ Explore possibilities
→ Develop direction
→ Receive conflicting feedback
```

```
→ Revise
→ Defend or adapt the work
```

Scenario Type E: Procedural and physical

Suitable for:

- Electrician
- Mechanic
- Chef
- Technician
- Machinist
- Clinical procedure roles

Typical rhythm:

```
Inspect environment
→ Identify tools and constraints
→ Sequence actions
→ Perform procedure
→ Detect error or hazard
→ Verify completion
```

Scenario Type F: Routine and endurance

Suitable for careers where consistency is central.

Typical rhythm:

```
Repeat structured tasks
→ Maintain attention
→ Manage gradual fatigue
→ Handle small variation
→ Preserve accuracy
→ Complete handoff
```

Why multiple rhythms are necessary

The structure of the simulation teaches users what kind of work rhythm they are experiencing. Forcing all careers into escalation would misrepresent quiet, repetitive, or methodical occupations.

8. Hybrid Time System

Time should advance primarily through actions and simulated time costs.

Examples:

- Reading a document advances 10 simulated minutes.
- Walking to another department advances 5 minutes.
- Completing a report advances 30 minutes.
- A meeting begins after a specified simulated time.
- A delayed task may expire after several actions.

Optional pressure mode

Hard real-time deadlines should not be required for baseline assessment.

When real-time pressure is educationally important, the system should provide:

- Standard mode
- Extended-time mode
- Pause-compatible mode
- No-hard-timer mode
- Practice mode

The debrief must record which pacing mode was used and avoid comparing response speed across incompatible modes.

Why this resolves the fairness problem

Real-time pressure can be a genuine part of some jobs, but reading speed, language background, disability, interface familiarity, and motor ability can affect timer performance.

The platform should simulate urgency without treating raw reaction speed as a general career-fit signal.

What may still be measured

The system can observe:

- Whether the user identifies urgency
- Which task they prioritize
- Whether they seek help
- Whether they communicate delays
- Whether they notice deteriorating conditions

These are often more meaningful than whether they clicked within 45 seconds.

9. Structured Simulation Engine

The backend, not the language model, controls:

- Time
- Locations
- Objects
- Task states
- Event triggers
- Character knowledge
- Character relationships
- Available actions
- Consequences
- Professional rules
- Skill evidence
- Safety constraints

```
Player action
↓
Input moderation and intent parsing
↓
Backend validates action
↓
Rules engine calculates effects
↓
World state changes
↓
Relevant NPC context is assembled
↓
AI generates bounded dialogue or narration
↓
Output is validated
↓
Approved output reaches the user
↓
Evidence and telemetry are recorded
```

Why this architecture is required

Language models can:

- Forget events
- Contradict previous statements
- Invent procedures

- Leak hidden information
- Resolve problems unrealistically
- Produce inconsistent consequences

The model should perform dialogue and narration, but it should not independently decide how the professional world operates.

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10. Main Technical Components

A. World Engine

Controls:

- Current location
- Connected locations
- Interactable objects
- Environmental changes
- Character positions
- Available movement

```
{
  "locationId": "security_office",
  "connectedLocations": [
    "operations_center",
    "meeting_room",
    "break_room"
  ],
  "interactables": [
    "workstation",
    "alert_dashboard",
    "team_chat",
    "incident_board"
  ]
}
```

B. Professional Systems Engine

Provides reusable interfaces for:

- Email
- Messaging
- Calendars

- Documents
- Dashboards
- Task boards
- Records
- Reports
- Knowledge bases
- Equipment controls

The same underlying component can be adapted across careers.

An inbox component may represent:

- Parent emails
- Patient messages
- Client requests
- Internal company communication

Scalability reason

Reusable system components reduce the amount of custom application code required for each career.

C. Event Engine

Each event should define:

```
{
  "eventType": "production_incident",
  "triggerConditions": [],
  "deadline": 20,
  "priority": "high",
  "possibleEscalations": [],
  "relatedCharacters": [],
  "affectedSystems": [],
  "skillSignals": [],
  "resolutionConditions": [],
  "prohibitedOutcomes": [],
  "accuracyReferences": []
}
```

Events can:

- Trigger
- Wait
- Escalate
- Expire

- Combine
- Block another task
- Produce delayed effects
- Resolve quietly
- Remain incomplete at the end of a shift

Not every event requires escalation.

D. Consequence Ledger

Every meaningful action can create immediate and delayed consequences.

```
{
  "action": "delayed_customer_response",
  "immediateEffect": "maintained_focus_on_primary_task",
  "delayedEffects": [
    {
      "triggerTime": "14:30",
      "effect": "customer_requests_status_update"
    }
  ]
}
```

The consequence ledger supports:

- Persistence
- Replay
- Explanation
- Testing
- Debrief evidence
- Cross-session patterns

E. NPC Engine

Each NPC contains:

- Role
- Goals
- Knowledge
- Responsibilities
- Communication style
- Relationship state
- Current workload

- Emotional state
- Memory of important interactions

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11. NPC Knowledge Isolation

A `doesNotKnow` field alone is not sufficient.

Each NPC generation call must receive a separately scoped context.

Context assembly process

```
Full world state
↓
NPC access-control filter
↓
Relevant known facts only
↓
Relevant memory only
↓
Current goals and emotional state
↓
Permitted dialogue boundaries
↓
LLM generation
```

The language model generating one NPC's dialogue should not receive facts that NPC is not permitted to know.

Additional protections

Pre-generation redaction

Remove restricted facts before creating the prompt.

Fact identifiers

Important facts should have IDs and visibility rules.

```
{
  "factId": "payment_failure_report",
  "knownBy": ["customer", "support_agent"],
```

```
  "notKnownBy": ["manager", "engineer"]
}
```

Post-generation leakage check

Generated dialogue should be checked against restricted facts before being displayed.

Adversarial tests

Tests should intentionally attempt to cause NPCs to:

- Reveal hidden information
- Refer to unseen events
- Know private conversations
- Infer confidential facts without evidence

Why this is necessary

Language models can infer or repeat information they receive even when instructed not to. Information isolation must happen before generation, not only through prompt instructions.

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12. Bounded Generative AI Layer

AI may generate:

- Dialogue wording
- Environmental descriptions
- Character tone
- Follow-up questions
- Minor scene variation
- Low-risk secondary interactions

AI may not independently change:

- Professional procedures
- Safety requirements
- Credentialing facts
- Major consequence logic
- Core scenario objectives
- Skill mappings
- NPC knowledge permissions
- Legal or ethical rules

Structured generation contract

The model should return data through a controlled schema.

```
{
  "dialogue": "I need a brief update before the meeting.",
  "tone": "concerned",
  "referencedFactIds": ["incident_open", "meeting_soon"],
  "proposedMinorStateUpdates": [
    {
      "field": "npcMood",
      "value": "impatient"
    }
  ]
}
```

The backend may accept, reject, or modify proposed updates.

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13. AI Consistency and Regression Testing

AI consistency needs its own engineering workstream.

A. Golden-path tests

Each scenario should have tested sequences covering:

- Expected actions
- Common alternative actions
- Incorrect actions
- Delayed responses
- Requests for help
- Off-script questions

Expected state changes should be stored and compared after updates.

B. Invariant tests

Certain rules must never be violated.

Examples:

- An NPC cannot reference unknown information.

- A patient cannot recover because of an unrelated dialogue choice.
- A user cannot access a locked system without authorization.
- A safety procedure cannot be skipped without a defined consequence.
- The model cannot invent credentials or career requirements.

C. Output contradiction checks

Before output reaches the user, validate it against:

- Current world state
- Approved career facts
- NPC knowledge
- Scenario rules
- Prohibited claims
- Safety requirements

D. Replay testing

Saved action sequences should be replayed after:

- Model updates
- Prompt changes
- Career-pack revisions
- Engine changes

This identifies whether previously stable simulations have drifted.

E. Human review sampling

A percentage of anonymized or consented runs should be reviewed for:

- Occupational inaccuracies
- Stereotypes
- Inconsistent consequences
- Misleading debriefs
- Inappropriate AI responses

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14. Free-Text Input Safety

Open-ended input is valuable because users may ask questions that authors did not predict.

It also creates risks.

Input processing should include:

- Harmful-content detection
- Harassment detection
- Sexual-content safeguards
- Self-harm escalation handling where appropriate
- Attempts to manipulate the simulation
- Attempts to extract hidden system instructions
- Repeated off-task behavior

NPC response policy

NPCs should:

- Maintain professional boundaries
- Redirect irrelevant behavior
- Respond safely to harassment
- Avoid escalating abusive dialogue
- Never reward harmful conduct
- Preserve the educational purpose

Simulation consequences versus platform safety

An unprofessional action may produce an in-world consequence, such as loss of trust.

However, harmful content should not simply be treated as an interesting gameplay choice. Platform safety controls operate before the simulation response.

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15. Career Knowledge and Content Governance

Each career pack requires structured knowledge about:

- Typical tasks
- Common tools
- Work rhythm
- Entry-level responsibilities
- Repetitive work
- Common frustrations
- Physical demands
- Emotional demands
- Ethical constraints
- Safety requirements

- Common misconceptions
- Career variation across industries
- Education and credential pathways

SME review pipeline

Every career pack should pass through:

1. Occupational research

Gather multiple credible sources rather than relying on one job description.

2. Practitioner interviews

Include professionals from different:

- Organizations
- Seniority levels
- Work settings
- Geographic areas where relevant

3. Scenario writing

Translate occupational findings into systems, events, characters, and routines.

4. Technical implementation

Configure the career pack using existing engine components.

5. SME review

At least one qualified practitioner reviews:

- Accuracy
- Missing responsibilities
- Unrealistic consequences
- Dramatic exaggeration
- Common stereotypes

6. Bias review

Review whether the career is being represented through clichés.

Examples to avoid:

- Nurses shown only through emergencies

- Cybersecurity shown only through dramatic hacking
- Lawyers shown mainly in court
- Teachers shown only managing disruptive students
- Engineers shown as socially isolated
- Skilled trades shown as purely physical labor

7. Pilot testing

Practitioners and target users complete the simulation.

8. Versioning and renewal

Career packs should display:

- Version
- Last reviewed date
- Reviewing roles
- Source update date

Dispute handling

If experts disagree, the career pack should:

- Represent legitimate workplace variation
- Avoid presenting one organization's process as universal
- Label context such as industry, location, or employer type
- Record disputed content for further review

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16. Career-Pack Architecture

A career should be added primarily through configuration.

```
career:
  id: cybersecurity_analyst
  title: Junior Cybersecurity Analyst
  pack_version: 1.0
  review_status: approved

work_dimensions:
  ambiguity: high
  social_interaction: medium
  interruption_level: high
  repetitive_work: medium
```

```
  physical_activity: low
  concentration_depth: medium

scenario_rhythms:
  - high_pressure_operational
  - quiet_investigation
  - routine_monitoring

locations:
  - office
  - operations_center
  - meeting_room

systems:
  - email
  - team_chat
  - alert_dashboard
  - ticket_queue
  - knowledge_base

npc_archetypes:
  - manager
  - senior_analyst
  - employee
  - client

event_templates:
  - suspicious_login
  - phishing_report
  - false_positive
  - incomplete_handoff
  - routine_alert_review

skills:
  - analytical_reasoning
  - communication
  - prioritization
  - attention_to_detail
```

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17. Authoring Cost and Scalability

Career packs are configuration-driven, but they are not cheap YAML files.

A high-quality pack requires:

- Research
- SME interviews
- Systems design
- Scenario writing
- Consequence mapping
- NPC design
- Skill mapping
- Testing
- Accessibility review
- Bias review
- Updates

Career complexity tiers

Tier 1: Primarily cognitive or social

Can reuse most existing interfaces.

Examples:

- Analyst
- Project coordinator
- Recruiter
- Teacher
- Customer success

Tier 2: Mixed cognitive and procedural

Requires some specialized systems.

Examples:

- Nurse
- Laboratory technician
- Construction manager
- Financial auditor

Tier 3: Highly physical or spatial

Requires custom interactive modules.

Examples:

- Electrician
- Mechanic

- Surgeon
- Pilot
- Machinist

Authoring estimates

Before committing to scale, the team should measure:

- Research hours per career
- SME review hours
- Scenario-authoring hours
- Technical implementation hours
- Testing hours
- Update cost per year

The first three packs should be used to establish realistic production estimates.

Why this matters

A reusable engine reduces software duplication. It does not eliminate the cost of producing accurate occupational content.

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18. Evidence and Skill Inference

The platform should combine three forms of evidence.

A. Behavioral evidence

What the user did.

B. Outcome evidence

What resulted from the action.

C. Reflective evidence

How the user experienced the work.

Example:

A user may successfully calm an angry customer but report that the experience was extremely draining.

That may indicate:

- Demonstrated communication capability
- Low preference for repeated customer conflict

Competence and enjoyment must remain separate.

Evidence categories

The engine should distinguish:

```
Demonstrated behavior
Current performance
Reported enjoyment
Reported energy impact
Reported confidence
Repeated preference
Values alignment
Interest in further exploration
```

No fake precision

During early phases, do not use scores such as:

```
Communication strength: 0.73
Career compatibility: 82%
```

Use confidence-aware categories:

- Insufficient evidence
- Early signal
- Repeated signal
- Strong repeated pattern

Why

Two or three observations do not justify decimal-level precision.

Numerical scoring may be used internally for processing, but user-facing language must reflect the limited evidence.

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19. Longitudinal Evidence Model

A single simulation should produce only provisional observations.

Confidence should increase only when patterns repeat across:

- Multiple events
- Multiple shifts
- Different scenarios
- Related careers
- Different interaction modes
- Self-reflection over time

Example:

```
Session 1:
Early signal of preference for independent investigation

Session 2:
Repeated preference for uninterrupted analytical work

Session 3:
Similar pattern in a different career context

Current interpretation:
Repeated evidence of preference for sustained individual focus
```

Contradictory evidence

The system must preserve contradictions.

Example:

- User enjoyed investigating a security incident.
- User disliked routine monitoring.
- User enjoyed collaboration in a design simulation.
- User disliked constant client interaction.

The platform should not force these into a single broad trait.

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20. Cross-Session Memory

The platform should maintain a longitudinal profile containing:

- Completed simulations
- Evidence records
- User reflections
- Confidence level
- Contradictory observations
- Careers explored
- Suggested next simulations
- User corrections

Users should be able to:

- Review stored observations
- Correct interpretations
- Delete individual sessions
- Reset their profile
- Control whether counselors or parents can see information

Why user correction matters

The system observes behavior but may misunderstand intent.

For example, a user may delay responding because they did not understand the interface, not because they avoid communication.

The user should be able to annotate:

> “I missed this because I did not notice the notification.”

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21. Debrief Design

The debrief should present:

A. What happened

A concise reconstruction of important events.

B. What the user did

Specific actions without moral judgment.

C. What consequences followed

Immediate and delayed outcomes.

D. Early evidence

Patterns supported by the session.

E. User experience

What felt:

- Energizing
- Draining
- Interesting
- Boring
- Stressful
- Meaningful

F. Confidence and limitations

Clearly state:

- How much evidence exists
- Which observations are provisional
- What the simulation did not measure

G. Career reality

Connect the experience to:

- Typical working conditions
- Career variations
- Education and credential requirements
- Entry barriers
- Compensation and market information
- Related careers

H. Next action

Provide a practical exploration step.

Example debrief

```
During this session, you repeatedly investigated before escalating.

Evidence:
- Reviewed authentication logs before contacting your manager
- Compared two alerts before closing either one
- Asked a colleague for missing context

Early observation:
You may prefer understanding a problem thoroughly before acting.

Potential tension:
In time-sensitive environments, you may need to communicate uncertainty earlier.

Your reflection:
You found the investigation engaging but the repeated interruptions draining.

Current interpretation:
This is an early signal that analytical work may interest you, while constant
live-response environments may be less appealing.

Evidence level:
Early signal based on one simulation. Additional scenarios are needed.
```

---

22. What Happens After the Debrief

The platform should guide the user into a next step.

Possible actions:

- Try another shift in the same career
- Explore a related career with a different work environment
- Compare two careers
- Complete a targeted skill activity
- Review training and credential pathways
- Watch or read verified practitioner accounts
- Prepare questions for an informational interview
- Find job-shadowing opportunities
- Connect with a school counselor or mentor
- Save a career to an exploration list

Example

A user likes cybersecurity investigation but dislikes constant interruptions.

The platform might suggest:

- Digital forensics
- Security research
- Threat intelligence
- Compliance analysis

It should explain why those roles may preserve some preferred elements while changing the work rhythm.

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23. Privacy, Consent, and Young Users

Because the platform may serve minors, privacy must be built into the product architecture.

Data minimization

Collect only what is needed for:

- Simulation state
- User-requested progress
- Evidence generation
- Research with appropriate consent

Avoid storing unnecessary raw conversations indefinitely.

Separate data categories

Store separately:

- Account information
- Simulation telemetry
- Reflections
- Research data
- Counselor-visible summaries

Consent controls

Users and guardians where required should understand:

- What is collected
- Why it is collected

- How long it is retained
- Who can access it
- Whether it is used for research
- Whether it is used to improve models

Reflection privacy

Statements about stress, confidence, enjoyment, or emotional response should not automatically be visible to:

- Parents
- Teachers
- Counselors
- Schools

Visibility should be explicit and configurable.

Deletion and export

Users should be able to:

- Delete sessions
- Delete reflections
- Export their data
- Reset longitudinal evidence
- Revoke sharing permissions

Institutional deployment

Before use in schools, the product should receive appropriate privacy, education-law, accessibility, and child-safety review. Legal counsel should verify applicable obligations rather than relying on the product team’s interpretation.

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24. Psychometric and Ethical Guardrails

The platform should be positioned as:

> An experiential career-exploration tool.

It should not be marketed as:

- A diagnostic assessment
- A hiring assessment
- A clinical instrument

- A guaranteed predictor of success
- A replacement for counseling

Prohibited uses during early phases

The system should not be used to:

- Reject users from educational opportunities
- Rank students
- Select employees
- Determine academic placement
- Make high-stakes recommendations without human review

Fairness research

Before broad institutional use, examine whether outcomes differ unfairly across:

- Language backgrounds
- Disability status
- Gaming experience
- Socioeconomic context
- Device quality
- Cultural communication styles
- Demographic groups where legally and ethically appropriate to study

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25. Cost and Infrastructure Model

The simulation should not make an LLM call for every minor interaction.

Use deterministic interactions for:

- Opening doors
- Viewing static documents
- Moving between rooms
- Standard system notifications
- Known task-state updates
- Repeated routine actions

Use AI generation for:

- Open-ended NPC questions
- Context-sensitive dialogue
- Reflection summarization
- Carefully bounded scenario variation

- Natural explanations

Cost controls

- Cache common dialogue patterns
- Pre-generate routine NPC responses
- Use smaller models for classification and moderation
- Use larger models only for complex dialogue
- Summarize history into structured memory
- Retrieve only relevant facts
- Limit generated response length
- Batch non-urgent analytics
- Track cost per completed session
- Establish a maximum generation budget per simulation

Latency target

The core interaction loop should remain responsive even when generative media is unavailable.

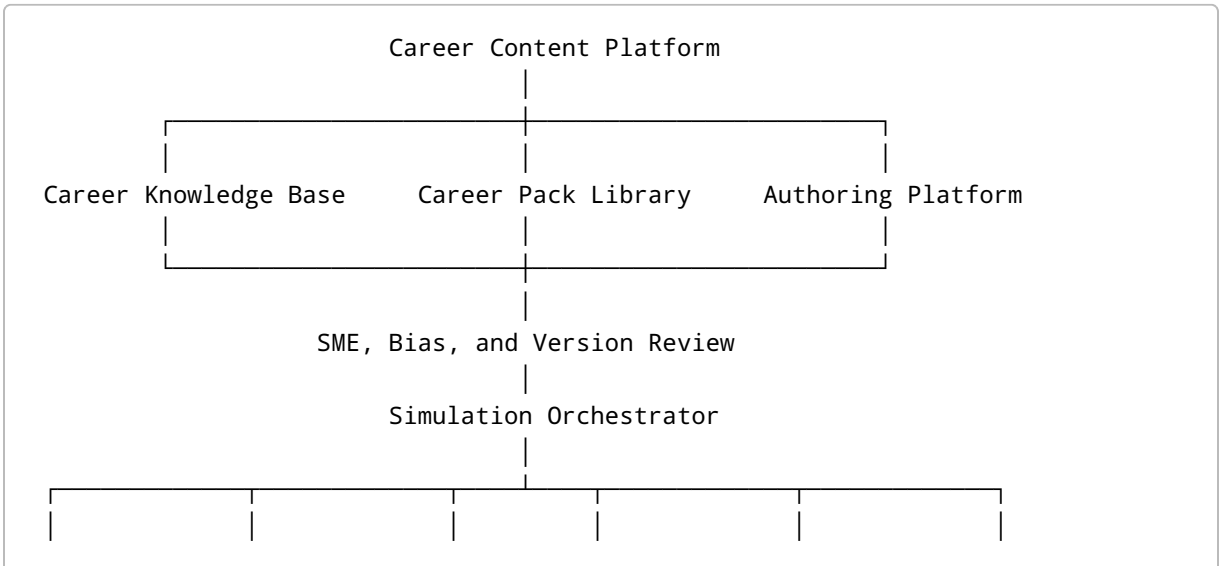
Scene images and voices should be:

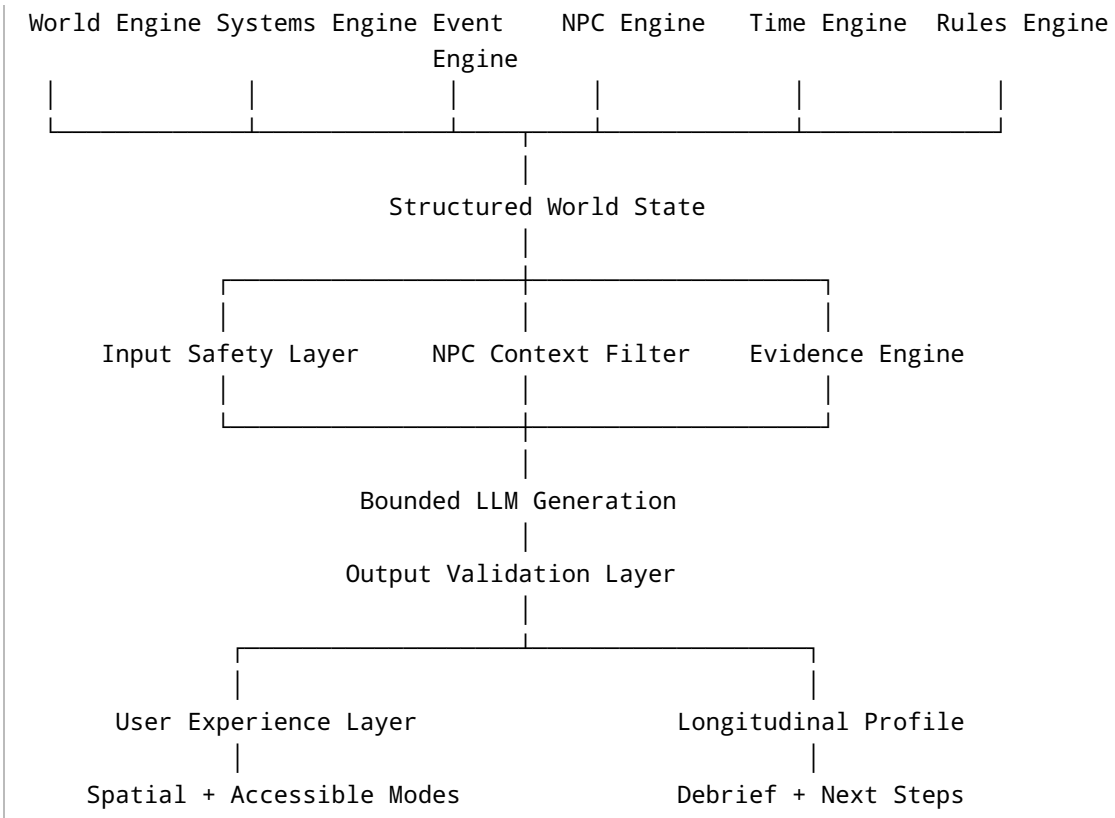
- Pre-generated
- Streamed
- Cached
- Optional

Waiting for image generation after every action would weaken immersion rather than improve it.

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26. Updated System Architecture





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27. Updated Development Roadmap

Phase 0: Research and claim definition

Before building a large engine:

- Define what the product is allowed to claim
- Define evidence categories
- Establish validation baselines
- Recruit initial SMEs
- Choose target user age group
- Define privacy requirements
- Define accessibility requirements
- Select one career for the vertical slice

Success condition

The team has a written claims policy and research plan before generating skill conclusions.

Phase 1: First validated vertical slice

Build one career with:

- One lightweight navigable workplace
- Three to five locations
- Three major NPCs
- Four to six professional systems
- Eight to twelve events
- At least two scenario rhythms
- One structured work shift
- Accessible parallel mode
- Tutorial
- Consequence ledger
- AI context isolation
- Input moderation
- Output validation
- Provisional debrief
- SME review
- User testing

Phase 1 success criteria

Measure separately:

Immersion

- Did users feel present?
- Did the world react consistently?

Occupational accuracy

- Did practitioners consider it representative?
- Did the simulation avoid stereotypes?

Understanding

- Can users explain the career more accurately afterward?

Decision usefulness

- Can users identify specific aspects they want to explore further?

Technical reliability

- Did NPCs leak information?
- Did AI contradict procedures?
- Did state remain consistent?

The phase does not claim to predict career success.

Phase 1.5: Validation checkpoint

Before scaling to more careers:

- Review evidence quality
- Remove misleading debrief claims
- Study interface confounds
- Compare against baseline career materials
- Test accessibility modes
- Measure session cost
- Estimate career-pack production effort
- Revisit privacy controls

Phase 2 should not begin merely because users enjoyed Phase 1.

Phase 2: Reusable career engine

Extract reusable components:

- Location templates
- Professional-system interfaces
- NPC archetypes
- Event templates
- Scenario-rhythm templates
- Career-pack format
- Evidence taxonomy
- Authoring tools
- Automated validation tests
- Content-review workflow

Add two substantially different careers.

Recommended combination:

1. Analytical or quiet-focus role
2. Social or educational role
3. Operational role from Phase 1

Reason

This tests whether the engine supports different work rhythms rather than only high-pressure office scenarios.

Phase 3: Longitudinal exploration

Add:

- Multiple shifts per career
- Cross-session memory
- Contradictory evidence handling
- Confidence progression
- Career comparison
- Next-step recommendations
- Pathway information
- User corrections
- Counselor-sharing controls

Only at this phase should the platform begin presenting repeated patterns with higher confidence.

Phase 4: Sensory immersion

Add selectively:

- Ambient audio
- Character voices
- Scene transitions
- Environmental animation
- Object inspection
- More detailed navigation
- Pre-generated expressions
- Limited real-time media

Media is introduced after occupational accuracy and decision usefulness are demonstrated.

Phase 5: Flagship physical-career modules

For physical careers, add specialized interaction layers:

- Spatial inspection
- Hazard recognition
- Tool selection
- Procedure sequencing
- Timing tasks
- Equipment interfaces
- Limited 3D or VR environments

The shared engine continues to manage:

- Clients
- Coworkers
- Deadlines
- Documentation
- Safety consequences
- Emotional and cognitive demands

Custom physical interaction should be added only where embodiment materially changes career understanding.

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28. Recommended First Career

Junior Cybersecurity Analyst

Environment

- Personal workstation
- Security operations room
- Meeting room
- Break room

Systems

- Alert dashboard
- Team chat
- Ticket queue
- Email
- Internal knowledge base
- Calendar

Scenario rhythms

Routine monitoring

- Review alerts
- Identify false positives
- Document work
- Maintain attention

Quiet investigation

- Gather evidence
- Search logs
- Compare unusual behavior

- Ask for context

Operational pressure

- Handle a credible incident
- Communicate incomplete information
- Decide whether to escalate
- Coordinate with others

Why this remains a strong first vertical slice

It provides:

- Multiple reusable professional systems
- Spatial movement
- Routine work
- Focused investigation
- Optional pressure
- NPC interaction
- Delayed consequences
- Clear information boundaries
- Strong testing opportunities for AI consistency

The simulation must avoid portraying cybersecurity only as dramatic hacking.

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29. What the MVP Should Not Include

Do not begin with:

- A massive open world
- Full avatar customization
- Multiplayer
- Hundreds of careers
- Fully procedural career generation
- Real-time images after every action
- Universal compatibility percentages
- Visible psychological scoring
- Mandatory reaction-speed tests
- High-stakes school recommendations
- Unrestricted AI control of the world
- Publicly shared reflection data
- Detailed physical simulations before the common engine is validated

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30. Final Architecture Principle

The complete product model is:

```
A navigable workplace creates presence.

Professional systems create authenticity.

Career-configurable rhythms create occupational truth.

Time and competing demands reveal work style.

Routine and tedium prevent fantasy.

Persistent consequences create seriousness.

Characters create social texture.

Structured state creates consistency.

Context isolation prevents knowledge leakage.

AI validation prevents procedural drift.

Accessible modes separate interface skill from career behavior.

Longitudinal evidence prevents shallow conclusions.

SME and bias review protect occupational accuracy.

Privacy controls protect reflection and behavioral data.

Pathway information connects interest to real-world feasibility.

Research validation determines what the platform is allowed to claim.

Next steps turn insight into action.
```

The final product is not:

```
Fortnite graphics + career quiz
```

It is not:

```
Chatbot + workplace role-play
```

It is not:

```
A one-session psychological assessment
```

It is:

> A research-informed, safety-controlled, accessible career-simulation platform that lets users experience realistic work environments, understand how they respond to different forms of work, accumulate evidence across multiple experiences, and make better-informed decisions about what to explore next.